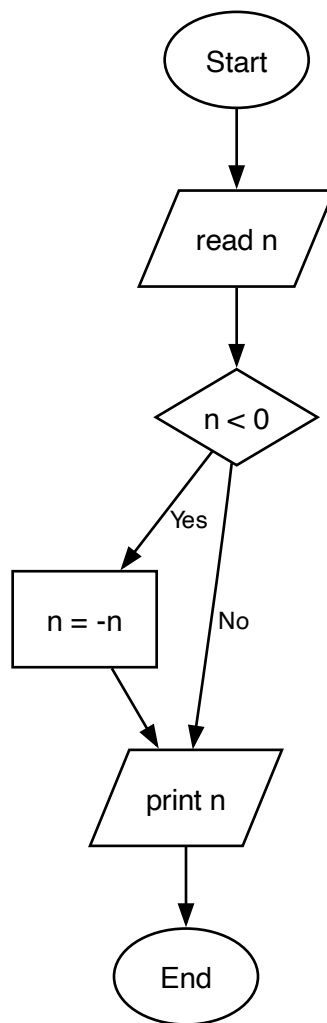


Answer key

Problem 1.



(i) For each input below, follow the flowchart and write the output.

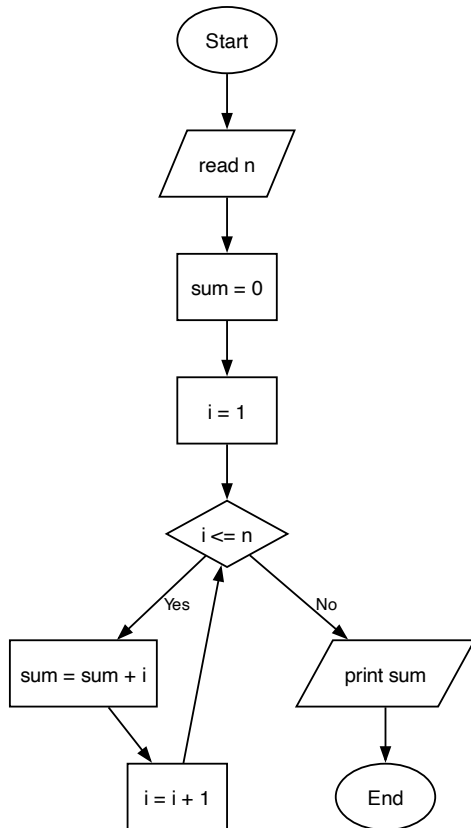
input n	output
7	7
-4	4
0	0
-19	19

(ii) Can you tell what this algorithm is doing?

Answer: Absolute value

Problem 2.

Trace this algorithm for $n = 4$, and complete the trace table.

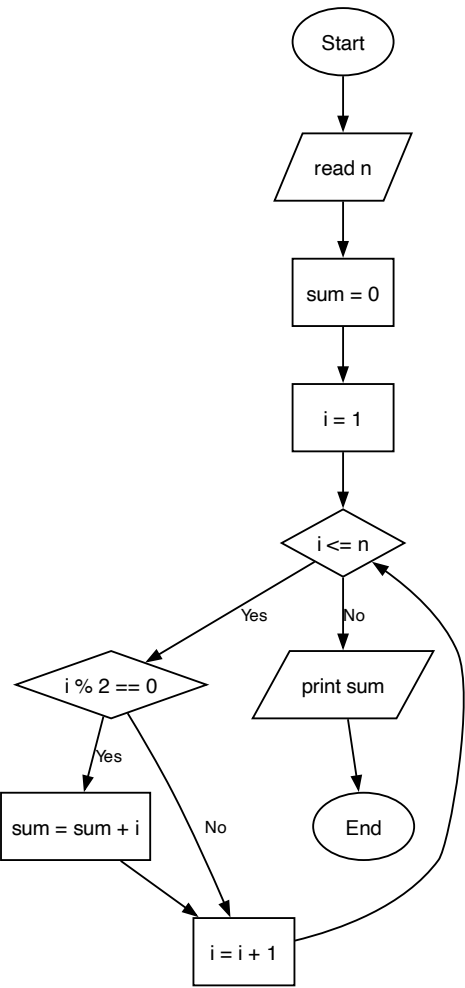


n	sum	i	output
4			
4	0		
4	0	1	
4	1	1	
4	1	2	
4	3	2	
4	3	3	
4	6	3	
4	6	4	
4	10	4	
4	10	5	
4	10	5	10

Output: 10

Problem 3.

Trace this algorithm for $n = 5$, and complete the trace table.

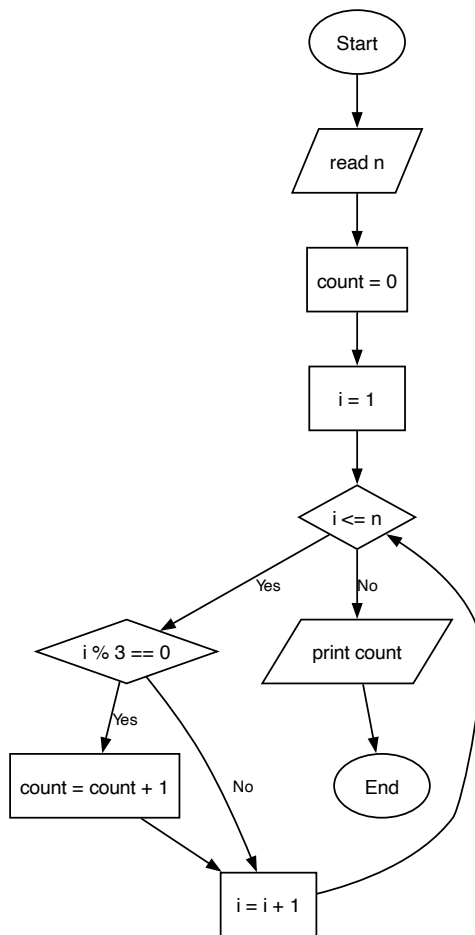


n	sum	i	output
5			
5	0		
5	0	1	
5	0	2	
5	2	2	
5	2	3	
5	2	4	
5	6	4	
5	6	5	
5	6	6	
5	6	6	6

Output: 6

Problem 4.

Trace this algorithm for $n = 7$, and complete the trace table.



n	count	i	output
7			
7	0		
7	0	1	
7	0	2	
7	0	3	
7	1	3	
7	1	4	
7	1	5	
7	1	6	
7	2	6	
7	2	7	
7	2	8	
7	2	8	2

Output: 2

Problem 5.

Trace this algorithm for $n = 472$, and complete the trace table.

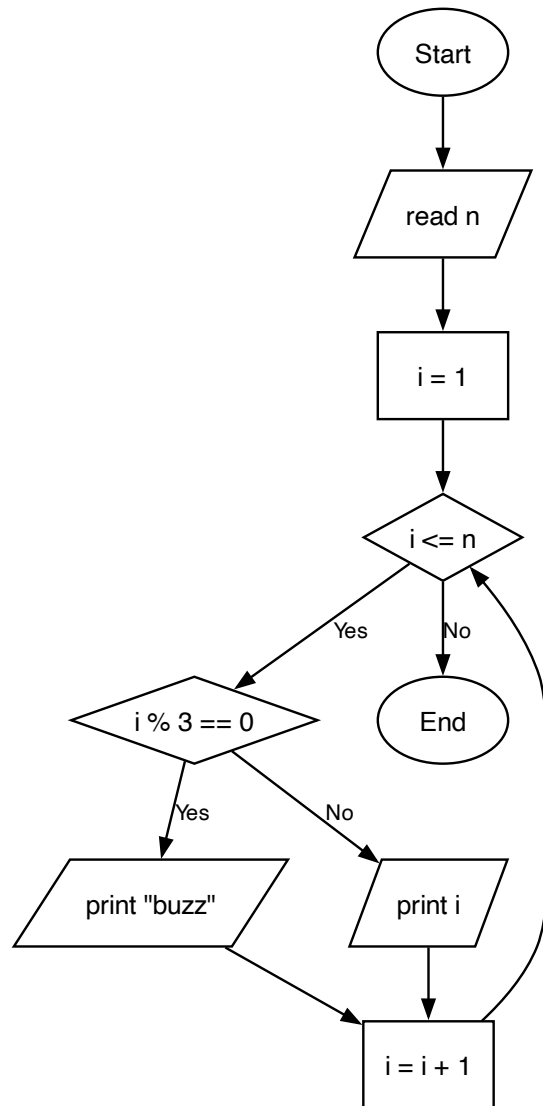
Note: $n // 10$ means whole-number division — divide and drop the remainder. For example, $472 // 10 = 47$ (not 47.2).



n	count	output
472		
472	0	
472	1	
47	1	
47	2	
4	2	
4	3	
0	3	
0	3	3

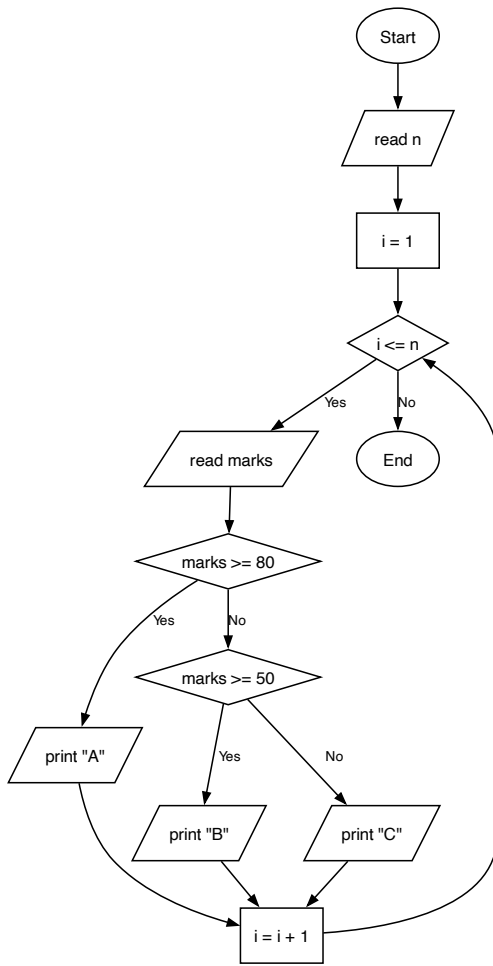
Output: 3

Problem 6.



Problem 7.

Trace this algorithm for $n = 3$, where the values of *marks* entered are 55, 82, 40 (in that order), and complete the trace table.



n	i	marks	output
3			
3	1		
3	1	55	
3	1	55	B
3	2	55	
3	2	82	
3	2	82	A
3	3	82	
3	3	40	
3	3	40	C
3	4	40	

Output: B, A, C